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Roll No. .....

324456(24)

B. E. (Fourth Semester) Examination, Nov.-Dec. 2015

(New Scheme)

(Elect. Branch)

ECTRICAL MEASUREMENTS & MEASURING INSTRUMENTS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

e : Attempt all the questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question. All questions carry equal marks.

Unit-I

Classify the resistances from the point of view of measurement. 2

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(b) Draw the circuit of a Wheatstone Bridge and derive the conditions of balance.

(c) Describe the working with the help of suitable diagrams, how (AC) potentiometer can be used for :

(i) Calibration of a voltmeter

(ii) Calibration of a wattmeter

(d) Write short notes on :

(i) Megger

(ii) Ohm-meter

Unit-II

2. (a) Describe the limitations of D'Sautys bridge and how are they overcome by using a modified form of D'Sautys bridge?

(b) Describe working of Heaviside Mutual Inductance bridge for determination of mutual inductance of a coil.

(c) An inductance of 0·22 H and 20Ω resistance is measured by comparison with a fixed standard inductance of 0.1 H and 40Ω resistance. They are

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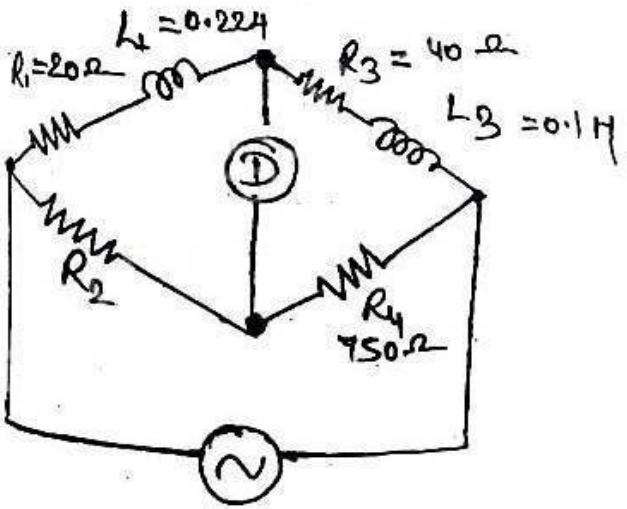
connected as shown in fig. The unknown inductance is in arm *ab*, and the standard inductance is arm *bc*, a resistance of 750Ω is connected in art *Cd* and a resistance whose amount is not known is in arm *da*. Find the resistance of arm *da* and show any necessary and practical additions required to active both resistive and inductive balance. 7



b) In an Aderson bridge for the measurement of inductance the arm *AB*, arm *BC* and arm *CD* and *DA* consist the elements as shown in figure.

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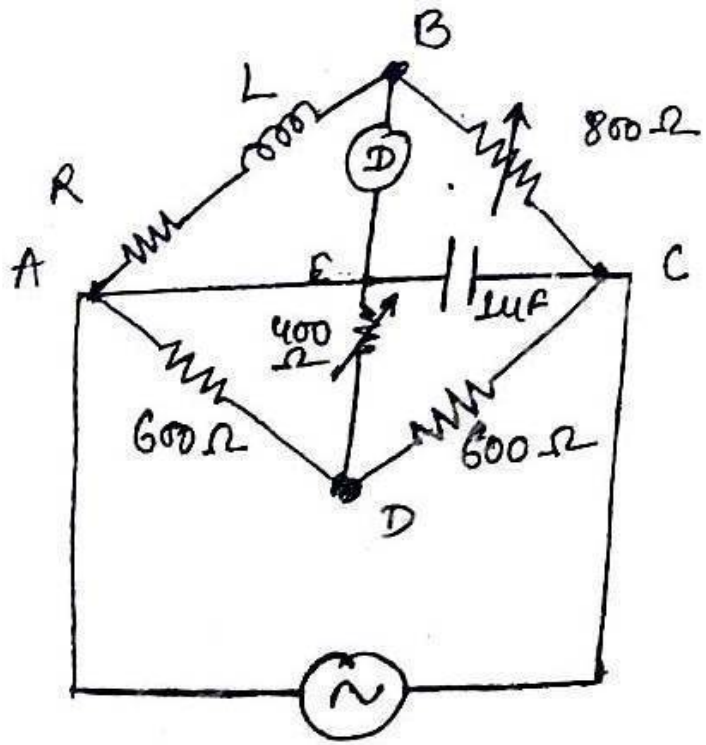

Calcuate the value of unknown *R* and *L*. Derive the condition for balance and draw the plusor diagram under balanced condition. 7

Unit-III

3. (a) Define the term "Galavanometer". 2

(b) Describe the principle and working of Ballistic galvanometer. Write their advantages and dis-advantages. 7

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(c) Explain the different parts of a CRT in brief.

(d) Explain the working of Lissajous patterns and draw the Lissajous patterns for different phase shift.

Unit-IV

4. (a) Explain the damping in the measuring instruments.

(b) Describe the working of Ratiometer type frequency meter.

(c) Write the comparison between attraction type instruments and Repulsion type instruments.

(d) Explain the working principle of MI instruments and prove that :

$$\theta = \frac{1}{2} \cdot \frac{I^2}{K} \cdot \frac{dL}{d\theta}$$

Unit-V

5. (a) What do you mean by "Creeping"?

(b) Explain the two wattmeter star (Wye) connection and delta connection for measuring the power.

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(c) Describe the construction of Induction type energy meters. 7

(d) Two wattmeters connected to measure the input to a balanced 3-phase circuit indicate 2000 W and 500 W respectively. Find the power factor of circuit.

(i) When both the readings are positive.

(ii) When both the latter reading is obtained after reversing the connections to the circuit coil of first instrument. 7

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